

Mathematical Analysis

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Source Code & Contributions

GitHub: <https://github.com/sihyeon8240/math>

Feel free to open issues or pull requests for corrections.

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Preface

This textbook uses Lean 4 to formalize selected mathematical statements and proofs.

- **Tools.** Lean is an interactive theorem prover; Mathlib is a library of formalized mathematics. This edition uses Lean 4.34.0 and Mathlib v4.34.0 (revision [5ed296525643](#)).
- **Reading the proofs.** Each paired statement or proof step gives a mathematical explanation followed by its Lean code. The explanations can be read independently; code fragments are read in order within the complete declaration.
- **Verification.** Lean checks formal proofs under their stated assumptions. Verification covers only statements linked to checked declarations. Agreement between the explanations, displayed code, and checked proofs is reviewed separately.

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1.3 Connected Sets

Chapter 2

Sequences and Series

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2.2 Subsequences

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The Riemann-Stieltjes Integral

5.1 Definition and Existence of the Integral

5.2 Properties of the Integral

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5.5 Rectifiable Curves

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Sequences and Series of Functions

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6.2 Uniform Convergence

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Functions of Several Variables

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Integration of Differential Forms

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9.2 Primitive Mappings

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9.4 Change of Variables

9.5 Differential Forms

9.6 Simplexes and Chains

9.7 Stokes' Theorem

9.8 Closed Forms and Exact Forms

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The Lebesgue Theory

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10.2 Construction of the Lebesgue Measure

10.3 Measure Spaces

10.4 Measurable Functions

10.5 Simple Functions

10.6 Integration

10.7 Comparison with the Riemann Integral

10.8 Integration of Complex Functions

10.9 Functions of Class $\mathcal{L}^2$

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